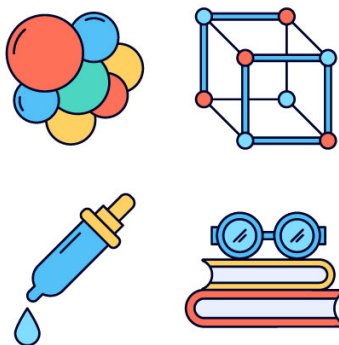




UNIVERSITY INTERSCHOLASTIC LEAGUE

Science

Invitational A • 2025



GENERAL DIRECTIONS:

- **DO NOT OPEN EXAM UNTIL TOLD TO DO SO.**
- Contestants may take up to two hours to complete the contest. If you are in the process of actually writing an answer when the signal to stop is given, you may finish writing that answer.
- Papers may not be turned in until 30 minutes have elapsed. If you finish the test in less than 30 minutes, remain at your seat and retain your paper until told to do otherwise. You may use this time to check your answers.
- All answers must be written on the answer sheet provided. Indicate your answers in the appropriate blanks provided on the answer sheet. Write clearly and legibly!
- You may place as many notations as you desire anywhere on the test paper but not on the answer sheet, which is reserved for answers only.
- You may use additional scratch paper provided by the contest director.
- All questions have ONE and only ONE correct (BEST) answer. There is a penalty for all incorrect answers.
- If a question is omitted, no points are given or subtracted.
- The back two pages of this test include a copy of the periodic table of the elements, as well as listings of other scientific relationships. You may use this information during the contest and may detach the back page from the test if you wish.
- A simple scientific calculator is sufficient for the high school Science contest. **The UIL provides a list of approved calculators that meet the criteria for use in the Science contest. No other calculators are permitted during the contest.** The Science Contest Approved Calculator List is available in the current Science Contest Handbook and on the UIL website. Contest directors will perform a brief visual inspection to confirm that all contestants are using only approved calculators. Each contestant may use up to two approved calculators during the contest.

- B01. Which of the following is not a member of the endomembrane system of the eukaryotic cell?
- A) rough endoplasmic reticulum
 - B) Golgi apparatus
 - C) nuclear envelope
 - D) mitochondria
 - E) smooth endoplasmic reticulum
- B02. Emesis directly affects the _____ system.
- A) integumentary
 - B) nervous
 - C) digestive
 - D) cardiovascular
 - E) skeletal
- B03. The ribosome is made of
- A) DNA and rRNA.
 - B) protein and tRNA.
 - C) DNA and protein.
 - D) rRNA and protein.
 - E) mRNA and rRNA.
- B04. Smooth, cardiac, and skeletal are three types of _____ tissue.
- A) muscle
 - B) connective
 - C) nervous
 - D) epithelial
 - E) integumentary
- B05. Scarlet Macaws and Great Green Macaws have been historically pre-zygotically separated by tall mountain ranges with cooler weather. Due to climate change, the weather in the mountains is warmer, which has provided an avenue for Scarlet Macaws to move across the barrier and mate with Great Green Macaws. What is a possible outcome of this hybridization?
- A) Increase in genetic variation
 - B) Increase in speciation
 - C) Wasted reproductive effort and resources
 - D) Extinction
 - E) All of the above are possible outcomes.
- B06. Malaria is caused by species belonging to Genus _____ and is spread by mosquitoes of Genus _____
- A) *Trypanosoma*; *Aedes*.
 - B) *Plasmodium*; *Anopheles*.
 - C) *Plasmodium*; *Aedes*.
 - D) *Trypanosoma*; *Anopheles*.
 - E) *Plasmodium*; *Culex*.
- B07. DNA replication in eukaryotes occurs within the
- A) nucleus.
 - B) nucleolus.
 - C) cytoplasm.
 - D) nucleoid.
 - E) ribosome.
- B08. From the human perspective, lice are
- A) endoparasites.
 - B) ectoparasites.
 - C) commensals.
 - D) mutualistic.
 - E) biological vectors.
- B09. The majority of biochemical reactions of cellular respiration occur
- A) in the nucleus.
 - B) outside the cell.
 - C) in the mitochondria.
 - D) in the plasma membrane.
 - E) within the endoplasmic reticulum.
- B10. Which major macromolecular group contains molecules that store energy in the long-term?
- A) Proteins
 - B) Nucleic acids
 - C) Carbohydrates
 - D) Amino acids
 - E) Lipids

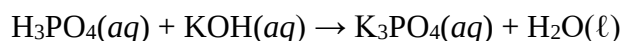
- B11. In a population at Hardy-Weinberg equilibrium, which value in the mathematical formula(s) represents the frequency of homozygous recessive individuals in the population?
- A) p^2
 - B) p
 - C) q^2
 - D) q
 - E) $2pq$
- B12. What major event is occurring in cytokinesis of the eukaryotic cell cycle?
- A) Condensation of the chromosomes
 - B) DNA replication
 - C) Cellular growth
 - D) Formation of the spindle apparatus
 - E) Division of the cytoplasm
- B13. Which of the following is an effect of not regulating gene expression?
- A) Upregulation of protein synthesis
 - B) Downregulation of protein synthesis
 - C) Prevention of gene expression
 - D) Cancer
 - E) All of the above are effects of not regulating gene expression.
- B14. *Jacana spinosa* is a bird native to Costa Rica. During the breeding season, one female *Jacana* sp. mates with multiple males and maintains several different nesting sites, which are incubated by the males. This type of mating system is specifically called
- A) monogamy.
 - B) polygyny.
 - C) polygynandry.
 - D) polyandry.
 - E) polygamy.
- B15. The Centers for Disease Control and Prevention issued a Food Safety Alert in October 2024 for McDonald's onions due to possible contamination with
- A) *Escherichia coli* O27.
 - B) *Escherichia coli* O157:H7.
 - C) *Salmonella*.
 - D) *Listeria*.
 - E) Measles.
- B16. *Homo sapiens* and *Homo habilis* belong to all of the following except
- A) Order
 - B) Family
 - C) Kingdom
 - D) Species
 - E) Genus
- B17. Which technology would be used to amplify a DNA target sequence on the genome?
- A) PCR
 - B) DNA gel electrophoresis
 - C) Northern Blot
 - D) Southern Blot
 - E) Western Blot
- B18. Consider the F₂ generation of Mendel's pea plants. If 16 progeny were produced in this generation from a dihybrid cross, statistically, how many of the 16 would be homozygous dominant for both traits?
- A) 1
 - B) 3
 - C) 6
 - D) 8
 - E) 16

- B19. The ornate and often colorful patterns on male birds is most derived from
- A) genetic drift.
 - B) sexual selection by females.
 - C) abundance of resources.
 - D) macroevolution.
 - E) environmental factors.
- B20. Protein synthesis is to _____ as cellular respiration is to _____.
- A) anabolism; anabolism
 - B) catabolism; catabolism
 - C) anabolism; catabolism
 - D) catabolism; anabolism

C01. What is the molar mass of K_2CO_3 ?

- A) 39.10 g/mol
- B) 67.11 g/mol
- C) 106.21 g/mol
- D) 126.20 g/mol
- E) 138.21 g/mol

C02. What is the sum of the coefficients when this reaction is balanced using the smallest whole number coefficients?



- A) 4 B) 6 C) 8 D) 10 E) 12

C03. Which of these statements is true when an electron in an oxygen atom falls from $n = 5$ to $n = 2$?

- A) Electricity is given off.
- B) Light is emitted.
- C) Sound is produced.
- D) Another electron goes from $n = 2$ to $n = 5$.
- E) The charge on the atom decreases.

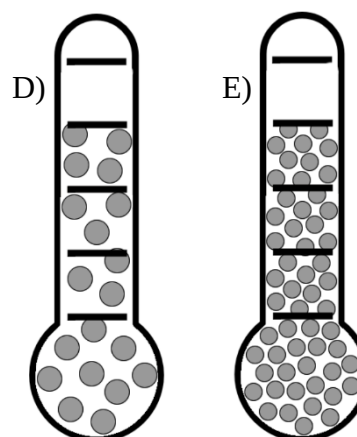
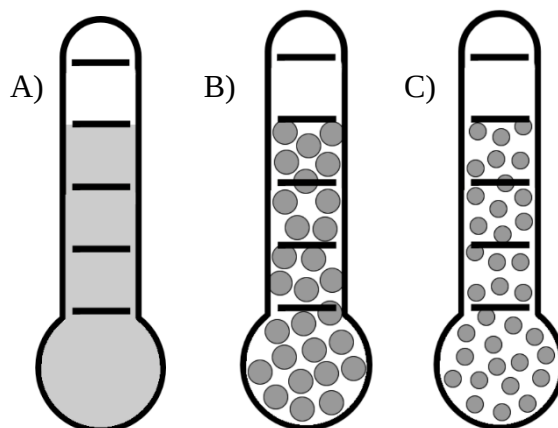
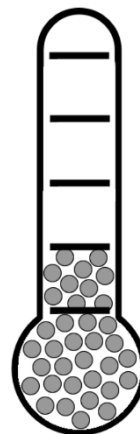
C04. According to valence bond theory, which of these is an example of a hybrid atomic orbital used in forming a covalent bond?

- A) sp^3 B) s C) s^2p D) p^2 E) π

C05. A mole of gas at STP takes up 22.4 liters. How hot would you have to heat the gas to make it take up twice that volume at 1 atm?

- A) 0°C
- B) 25.0°C
- C) 44.8°C
- D) 100°C
- E) 273°C

C06. This image shows a thermometer at a low temperature. Which of the following images best represents the same thermometer at a higher temperature?



C07. If you have 100 mL of 0.10 M HCl and you add 100 mL of water, what is the new pH of the solution?

- A) 1.0 B) 1.3 C) 0.5 D) 2.0 E) 2.3

C08. How much heat is given off when 100.0 grams of propanol ($\text{C}_3\text{H}_8\text{O}$) is burned? The heat of combustion for propanol is -2021 kJ/mol .

- A) 20.21 kJ B) 2.021 kJ
C) 3362 kJ D) 5600 kJ
E) $1.215 \times 10^5 \text{ J}$

C09. An aluminum cylinder at a temperature of 88.0°C is dropped into 125 grams of water at a temperature of 25.0°C . If the final temperature of the water and the aluminum is 25.8°C , what is the mass of the aluminum cylinder? $c_{\text{Al}} = 0.897 \text{ J/g}^\circ\text{C}$

- A) 7.5 g B) 8.0 g C) 8.5 g
D) 9.0 g E) 9.5 g

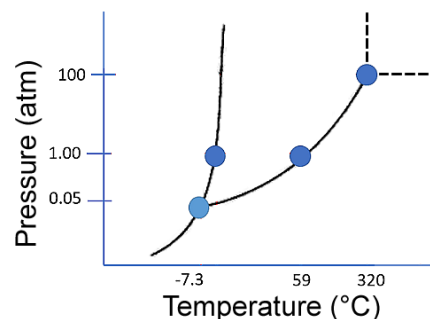
C10. Which of these determines the ratio of products to reactants in an equilibrium reaction that has reached equilibrium?

- A) The amount of reactants you start with
B) The amount of products present in the starting mixture
C) The strength of the intermolecular forces between the reactant molecules and the product molecules.
D) The speeds of the forward and backward reactions.
E) At equilibrium the ratio of products to reactants always equals 1.

C11. Which combination of aqueous reagents below will result in the formation of a solid precipitate?

- A) $\text{AgNO}_3 + \text{NaCl}$
B) $\text{NaNO}_3 + \text{Pb}(\text{NO}_3)_2$
C) $\text{NaCl} + \text{KBr}$
D) $\text{NaHCO}_3 + \text{NaOH}$
E) All of these

C12. This is an unlabeled phase diagram for an unknown compound. What physical state is the compound in under ordinary laboratory conditions?



- A) Bose-Einstein Condensate
B) Solid
C) Liquid
D) Gas
E) Supercritical Fluid

C13. What type of attraction holds neighboring molecules together in a covalent liquid?

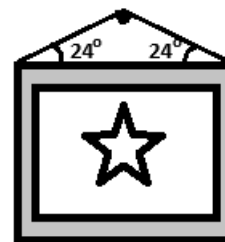
- A) Covalent bonds
B) Ionic bonds
C) Magnetic attraction forces
D) Intermolecular forces
E) Delta Forces

C14. Propane gas (C_3H_8) is completely combusted using excess O_2 gas. What are the products of the combustion reaction?

- A) CO_2
B) CO_2 and H_2O
C) CO_2 and CO
D) CO and H_2O
E) CO_2 and CH_4

- C15. What is the difference between the formal charge and the oxidation number on an atom in a compound?
- A) Nothing. The oxidation number of an atom in a compound is the same as the formal charge on that atom.
 - B) Formal charges are always the actual ionic charge on the atom and oxidation numbers are not.
 - C) Oxidation numbers are always the actual ionic charge on the atom and formal charges are not.
 - D) Formal charges are based on bonding electrons being shared equally and oxidation numbers are based on no shared electrons.
 - E) Oxidation numbers are based on bonding electrons being shared equally and formal charges are based on no shared electrons.
- C16. How many mL of water would you have to add to 75.0 mL of a 5.00 M stock solution in order to create a solution with a concentration of 4.00 M?
- A) 18.75 mL
 - B) 25.0 mL
 - C) 93.75 mL
 - D) 15.0 mL
 - E) 60.0 mL
- C17. Which of these is not a real covalent compound name?
- A) nitrogen triiodide
 - B) sulfur hexafluoride
 - C) dinitrogen pentoxide
 - D) carbon tetrachloride
 - E) monophosphorus monoxide
- C18. Aluminum fluoride, AlF_3 , is usually made by fluorinating aluminum metal using HF gas. How many grams of aluminum fluoride could you make from 225 grams of solid aluminum and excess HF gas?
- A) 700 g
 - B) 675 g
 - C) 525 g
 - D) 450 g
 - E) 225 g
- C19. How many atoms are there in a 16.0 L sample of CO_2 gas at 79 °C and 1.0 atm pressure?
- A) 3.34×10^{23} atoms
 - B) 1.00×10^{24} atoms
 - C) 4.05×10^{25} atoms
 - D) 6.02×10^{23} atoms
 - E) 6.64×10^{23} atoms
- C20. You are trying to carry out a reaction between a solid reactant and an aqueous reactant, but the reaction is going too slow. Which of these changes would speed up the reaction?
- A) heat the solution
 - B) chop up the solid reactant into smaller pieces
 - C) add more solid reactant
 - D) increase the concentration of the aqueous reactant
 - E) all of these

- P01. According to Orzel, observers moving at constant speed relative to each other define a special group of....
 A) velocity transforms
 B) reference transforms
 C) inertial frames
 D) relative frames
 E) acceleration frames
- P02. According to Orzel, Maxwell's equations translate into four simple rules. Which of these choices is NOT one of the rules derived from Maxwell's Equations?
 A) The strength of an electric field depends on the amount of charge in the vicinity.
 B) The strength of a magnetic field depends on the number of poles in the vicinity.
 C) A magnet will always have both north and south poles.
 D) An electric field is created by a changing magnetic field.
 E) A magnetic field is created by either a current or a changing electric field.
- P03. According to Orzel, the experiment to look for the effects of motion on the speed of light was carried out by....
 A) Poincare and Einstein
 B) Hertz and Lorentz
 C) FitzGerald and Lorentz
 D) Michaelson and Hertz
 E) Michaelson and Morley
- P04. The atmosphere of Venus is primarily composed of what gas?
 A) Nitrogen
 B) Oxygen
 C) Carbon Dioxide
 D) Sulfur Dioxide
 E) Carbon Monoxide
- P05. A marathon runner travels 1500 feet in 2.8 minutes. What is the runner's speed in meters per second?
 A) 2.7 m/s
 B) 3.8 m/s
 C) 5.4 m/s
 D) 7.6 m/s
 E) 8.9 m/s
- P06. You are resting on the ground when an angry fairy throws an acorn at you. The acorn is thrown straight down from a height of 4.40m above you, and with an initial velocity of 8.60m/s. How fast is the acorn moving when it hits you?
 A) 8.60 m/s
 B) 10.8 m/s
 C) 12.7 m/s
 D) 15.2 m/s
 E) 17.9 m/s
- P07. A 1.20kg framed picture is hung on a wall using a single wire looped over a nail, as shown. The wire makes an angle of 24.0° with respect to horizontal on both sides of the frame. What is the tension in the wire?
 A) 12.0 N
 B) 14.4 N
 C) 19.2 N
 D) 24.1 N
 E) 28.9 N



- P08. After being hit with an acorn, you angrily throw it directly upwards. The mass of the acorn is 2.50g and you throw it with an initial kinetic energy of 0.180J. Assuming it doesn't hit anything, how high above you does the acorn go after you throw it?
 A) 2.94 m
 B) 4.40 m
 C) 6.00 m
 D) 7.35 m
 E) 12.0 m
- P09. A circular disk with a radius of 25.0cm starts from rest. The disk spins up with an angular acceleration of 30.0rad/s^2 until it reaches a final angular velocity of 150rpm. How long does it take to reach this final angular velocity?
 A) 5.00 sec
 B) 2.10 sec
 C) 1.25 sec
 D) 0.524 sec
 E) 0.131 sec

P10. A 300.0g mass is attached to a vertical spring and released, after which the mass-spring system oscillates with a period of 0.855 seconds. What is the spring constant of the spring?

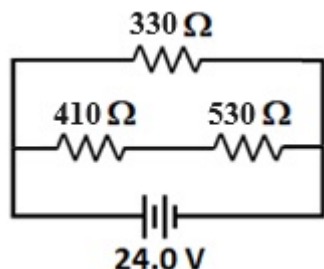
- A) 1.01 N/m
- B) 2.20 N/m
- C) 7.33 N/m
- D) 13.5 N/m
- E) 16.2 N/m

P11. A metal pipe is measured to have a length of 3.311m at a temperature of -15.0°C . At a temperature of $+42.0^{\circ}\text{C}$, the pipe is measured to be 3.324m in length. What is the coefficient of thermal expansion for the metal pipe?

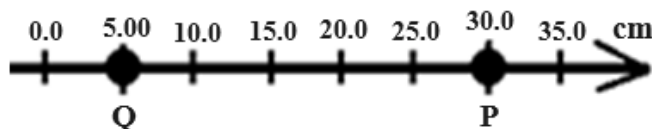
- A) $4.8 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$
- B) $2.3 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$
- C) $1.5 \times 10^{-4} \text{ }^{\circ}\text{C}^{-1}$
- D) $6.9 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$
- E) $4.4 \times 10^{-5} \text{ }^{\circ}\text{C}^{-1}$

P12. For the circuit shown below, what is the current flowing through the 530Ω resistor?

- A) 25.5 mA
- B) 45.3 mA
- C) 58.5 mA
- D) 72.7 mA
- E) 98.3 mA



P13. A charge, $Q = 33.0\text{nC}$, is placed on a horizontal axis as shown below. Find the magnitude of the electric potential at the point P (also on the axis) due to the charge Q.



- A) 990 V
- B) 1190 V
- C) 2960 V
- D) 4750 V
- E) 5930 V

P14. A fast-moving proton enters a region of magnetic field. The field strength is 450 Gauss, and the field is oriented perpendicular to the motion of the proton. If the proton traces out a circle with a diameter of 26.0cm, then what is the magnitude of the velocity of the proton?

- A) $3.50 \times 10^5 \text{ m/s}$
- B) $5.04 \times 10^5 \text{ m/s}$
- C) $5.60 \times 10^5 \text{ m/s}$
- D) $8.40 \times 10^5 \text{ m/s}$
- E) $1.12 \times 10^6 \text{ m/s}$

P15. A rectangular loop of wire is 30.0cm long, 15.0cm wide, and has a resistance of 1.20Ω . Initially, a 220 Gauss magnetic field passes through the loop, oriented perpendicular to the face of the loop. The magnetic field drops to zero in a time of $62.0\mu\text{s}$. What is the magnitude of the current induced in the loop due to the changing magnetic field?

- A) 16.0 A
- B) 13.3 A
- C) 8.87 A
- D) 4.44 A
- E) 2.96 A

P16. A moth with a wingspan of 5.00cm flies 56.0cm in front of a concave mirror. The mirror has a radius of curvature of 200.0cm. What is the wingspan of the image of the moth formed by the mirror?

- A) 3.21 cm
- B) 3.91 cm
- C) 6.94 cm
- D) 8.45 cm
- E) 11.4 cm

P17. Light with a wavelength of 455nm is directed onto a metal surface, resulting in electrons being emitted from the surface. The workfunction of the metal is 2.28eV. What is the velocity of the electrons that are emitted from the surface?

- A) $4.45 \times 10^4 \text{ m/s}$
- B) $1.57 \times 10^5 \text{ m/s}$
- C) $1.98 \times 10^5 \text{ m/s}$
- D) $2.73 \times 10^5 \text{ m/s}$
- E) $3.96 \times 10^5 \text{ m/s}$

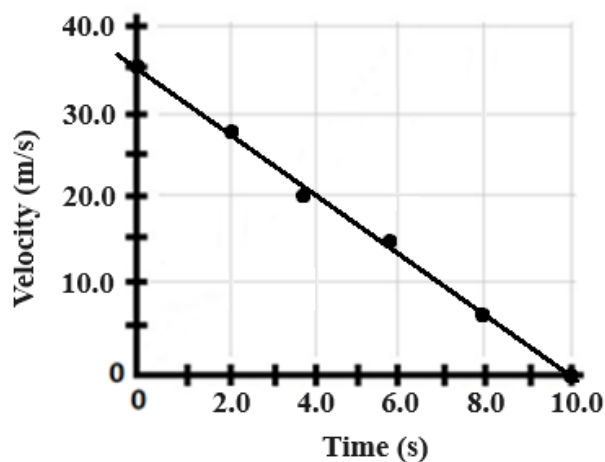
- P18. An atom of Copernicium ($^{285}_{112}\text{Cn}$) undergoes the following radioactive decays:

$$\alpha, \alpha, \gamma, \beta^-, \alpha, \gamma, \beta^-$$

What is the daughter isotope resulting from this series of radioactive decays?

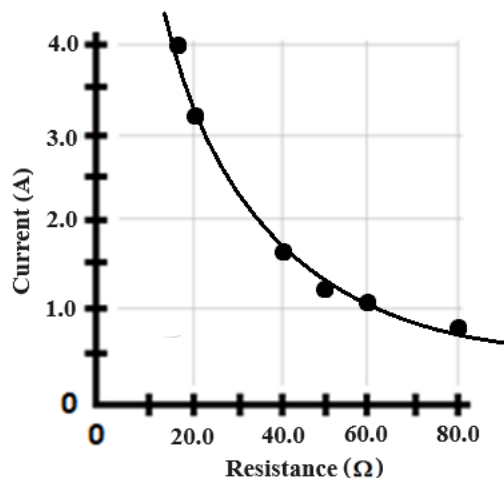
- A) $^{273}_{108}\text{Hs}$
- B) $^{275}_{108}\text{Hs}$
- C) $^{271}_{104}\text{Rf}$
- D) $^{273}_{104}\text{Rf}$
- E) $^{275}_{104}\text{Rf}$

- P19. The plot below illustrates the velocity of a small boat as a function of time. Based on these data, what is the acceleration of the boat?



- A) -1.5 m/s^2
- B) -2.3 m/s^2
- C) -3.5 m/s^2
- D) -4.4 m/s^2
- E) -5.8 m/s^2

- P20. You have a power supply that produces an unknown voltage. You connect the supply to different resistors and measure the current for each resistance. The data are plotted below. Based on these data, what is the magnitude of the voltage produced by the power supply?



- A) 25 V
- B) 40 V
- C) 50 V
- D) 65 V
- E) 100 V

Chemistry																		1A 1	2A 2											3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01			3 Li 6.94	4 Be 9.01													5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18													
11 Na 22.99	12 Mg 24.31	3B 3	4B 4	5B 5	6B 6	7B 7	8B 8 9 10			1B 11	2B 12	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95																		
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80																		
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.76	52 Te 127.60	53 I 126.90	54 Xe 131.29																		
55 Cs 132.91	56 Ba 137.33	57 La 138.9	72 Hf 178.49	73 Ta 180.95	74 W 183.84	75 Re 186.21	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.20	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)																		
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (281)	111 Rg (281)	112 Cn (285)	113 Nh (286)	114 Fl (289)	115 Mc (289)	116 Lv (293)	117 Ts (293)	118 Og (294)																		

58	Ce	140.1	59	Pr	140.9	60	Nd	144.2	61	Pm	(145)	62	Sm	150.4	63	Eu	152.0	64	Gd	157.3	65	Tb	158.9	66	Dy	162.5	67	Ho	164.9	68	Er	167.3	69	Tm	168.9	70	Yb	173.0	71	Lu	175.0
90	Th	232.0	91	Pa	231.0	92	U	238.0	93	Np	(237)	94	Pu	(244)	95	Am	(243)	96	Cm	(247)	97	Bk	(247)	98	Cf	(251)	99	Es	(252)	100	Fm	(257)	101	Md	(258)	102	No	(259)	103	Lr	(262)

Water Data

T_{mp}	=	0°C
T_{bp}	=	100°C
c_{ice}	=	2.09 J/g·K
c_{water}	=	4.184 J/g·K
c_{steam}	=	2.03 J/g·K
ΔH_{fus}	=	334 J/g
ΔH_{vap}	=	2260 J/g
K_{f}	=	1.86 °C/m
K_{b}	=	0.512 °C/m

Constants

R	=	0.08206 L·atm/mol·K
R	=	8.314 J/mol·K
R	=	62.36 L·torr/mol·K
e	=	1.602×10^{-19} C
N_{A}	=	6.022×10^{23} mol ⁻¹
k	=	1.38×10^{-23} J/K
h	=	6.626×10^{-34} J·s
c	=	3.00×10^8 m/s
R_{H}	=	2.178×10^{-18} J
m_{e}	=	9.11×10^{-31} kg
$\mathcal{F}$	=	96,485 C/mol e ⁻
1 amp	=	1 C/sec
1 mol e ⁻	=	96,485 C

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Physics

Useful Constants

quantity	symbol	value
Free-fall acceleration	g	9.80 m/s^2
Permittivity of Free Space	ϵ_0	$8.854 \times 10^{-12} \text{ C}^2/\text{Nm}^2$
Permeability of Free Space	μ_0	$4\pi \times 10^{-7} \text{ Tm/A}$
Coulomb constant	k	$8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$
Speed of light in a vacuum	c	$3.00 \times 10^8 \text{ m/s}$
Fundamental charge	e	$1.602 \times 10^{-19} \text{ C}$
Planck's constant	h	$6.626 \times 10^{-34} \text{ Js}$
Electron mass	m_e	$9.11 \times 10^{-31} \text{ kg}$
Proton mass	m_p	$1.67265 \times 10^{-27} \text{ kg}$ 1.007276 amu
Neutron mass	m_n	$1.67495 \times 10^{-27} \text{ kg}$ 1.008665 amu
Atomic Mass Unit	amu	$1.66 \times 10^{-27} \text{ kg}$ $931.5 \text{ MeV}/c^2$
Gravitational constant	G	$6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$
Stefan-Boltzmann constant	σ	$5.67 \times 10^{-8} \text{ W/m}^2\text{K}^4$
Universal gas constant	R	$8.314 \text{ J/mol} \cdot \text{K}$ $0.082057 \text{ L} \cdot \text{atm/mol} \cdot \text{K}$
Boltzmann's constant	k_B	$1.38 \times 10^{-23} \text{ J/K}$
Speed of Sound (at 20°C)	v	343 m/s
Avogadro's number	N_A	$6.022 \times 10^{23} \text{ atoms/mol}$
Electron Volts	eV	$1.602 \times 10^{-19} \text{ J/eV}$
Distance Conversion	miles → meters inches → centimeters	1.0 mile = 1609 meters 1.00 inch = 2.54 centimeters
Rydberg Constant	R_∞	$1.097 \times 10^7 \text{ m}^{-1}$
Standard Atmospheric Pressure	1 atm	$1.013 \times 10^5 \text{ Pa}$
Density of Pure Water	ρ_{water}	1000.0 kg/m^3
Magnetic Field Conversion	Gauss → Tesla	$10^4 \text{ Gauss} = 1.00 \text{ Tesla}$